

AB-620 — TWO CAPSTONE PROJECTS

Microsoft Certified: AI Agent Builder Associate

Designing and Building Integrated AI Agent Solutions in Microsoft Copilot Studio

Exam alignment

Associate (Developer) | 120 minutes | Passing score stated in supplied TOC: 700/1000

Domains: Plan & configure (30–35%); Integrate & extend (40–45%); Test & manage (20–25%).

These are training project briefs, not official exam questions or a certification guarantee. Use approved sample data and mock APIs when needed.

Capstone 1 — Enterprise Employee Service Desk: Multi-Agent AI Solution

Coordinator + IT + HR + Finance agents | Copilot Studio | Connected agents / A2A-ready design

1. Business scenario

Employees need one conversational entry point for onboarding, IT support, HR policy questions, and finance guidance. A coordinator agent delegates requests to specialist agents, grounds answers in approved knowledge, and provides next steps or human escalation. Transactional actions may be mocked unless authorized systems are available.

2. Business goals

- Provide one employee-facing entry point for HR, IT, and Finance.
- Delegate multi-domain requests to the right specialist agents and consolidate their responses.
- Ground answers in approved policies and documents; avoid unsupported claims.
- Demonstrate environment, DLP, security, ALM, testing, deployment, and monitoring practices.

3. Personas

- Employee — asks a question or requests help.
- Employee Service Coordinator — classifies, delegates, and summarizes.
- IT Support Agent — devices, VPN, Microsoft 365, access, incidents.
- HR Service Agent — onboarding, leave, benefits, policies.
- Finance Service Agent — expenses, reimbursements, invoices, procurement.
- Maker/service owner — configures access, tests, publishes, monitors.

4. Proposed architecture

- Employee in Teams/web → Employee Service Coordinator in Copilot Studio.
- Coordinator delegates to IT, HR, and Finance agents using supported connected-agent capabilities; use A2A where the chosen agent types and tenant support it.
- Specialists use approved SharePoint, website, document, or Dataverse knowledge; optionally Azure AI Search RAG with citations.
- One authorized agent flow or REST/custom connector action; use a mock endpoint if a real ticketing system is unavailable.
- Dev → Test/UAT → Production with solutions, connection references, environment variables, DLP, and least-privilege roles.

5. Functional and technical requirements

Area	Requirements
Agent behavior	Classify requests; delegate to one or more specialists; ask focused clarifying questions; consolidate answers; offer escalation; distinguish advice from completed actions.
Knowledge/RAG	Load sample HR, IT, and Finance policies/FAQs; test relevance, grounding, citations where supported, and unsupported questions.
Integration	Connect at least two specialists and implement one agent flow or REST action, real or mock. Handle errors and unavailable agents.
Security/governance	Define environments, DLP, security roles, connection approvals, data classification, and ALM. Do not disclose sensitive employee or payroll data.
Testing/operations	Test routing, ambiguity, privacy, failure, escalation, analytics, channel publishing, deployment, and rollback.

6. Implementation plan

Phase	Activities
1 — Plan & configure	Define use cases, blueprint, environments, DLP, roles, data boundaries, solution strategy, and ALM.
2 — Build agents	Create coordinator plus IT, HR, and Finance specialists; write instructions and delegation descriptions; configure fallback.
3 — Knowledge/RAG	Add approved policy files and a SharePoint/website source; optionally configure Azure AI Search RAG; test grounding.
4 — Integrate & extend	Connect agents; add one flow or REST/custom connector action; optionally add MCP; test hand-off, timeouts, and API errors.
5 — Test & manage	Run test matrix; review transcripts/analytics; publish to Teams/web; package solution and document promotion/rollback.
6 — Demo	Show a multi-department onboarding request, an IT-only request, and a failure/privacy scenario.

7. Demo / test scenarios

Scenario	Expected behavior
Multi-agent onboarding	"A new employee joins Monday. Explain joining documents, laptop/Microsoft 365 setup, and relocation expense steps." Expected: HR, IT, Finance are consulted and one consolidated plan is returned.
IT-only	"My VPN fails after a password reset." Expected: grounded IT troubleshooting and escalation guidance.
Knowledge missing	"What is the reimbursement deadline?" If no approved source contains it, say it cannot be verified; do not invent a deadline.

Scenario	Expected behavior
Tool failure	Request ticket creation while the tool is unavailable. Do not claim a ticket was created; provide the approved fallback.
Privacy	Ask for a colleague's salary or leave balance. Refuse unauthorized disclosure and point to the approved channel.

8. Deliverables

- Architecture and use-case blueprint.
- Four agents: coordinator, IT, HR, and Finance.
- Knowledge-source inventory and retrieval test results.
- At least one flow/REST action with success and failure behavior.
- Environment, DLP, security, connection, and ALM notes.
- Test matrix with expected/actual results and defects fixed.
- Deployment/monitoring notes and an 8–10 minute demo.

9. Acceptance criteria

- At least 8/10 core tests pass; all critical privacy/security tests pass.
- Coordinator delegates correctly for defined scenarios.
- No invented policies, approvals, ticket IDs, employee records, or completed actions.
- One integration path demonstrates validation and error handling.
- Deployment and rollback steps are documented.

10. AB-620 domain mapping

Exam domain	Project evidence
Plan & configure (30–35%)	Use cases, agent patterns, environment/DLP, governance, security roles, solution/ALM, connections/shared components.
Integrate & extend (40–45%)	Connected agents/A2A pattern, REST/custom connector, flow, optional MCP, RAG integration, error handling.
Test & manage (20–25%)	Test pane, conversation validation, analytics, Teams/web channel, solution deployment, monitoring/maintenance.

Suggested guided build: 3–5 hours, with optional extension time. This is a training estimate, not an exam requirement.

Suggested 10-minute presentation

- 1 min: problem, users, success measures.
- 2 min: architecture, agent roles, knowledge, and governance.
- 4 min: end-to-end demo including a failure/escalation.
- 2 min: tests, analytics, deployment, lessons learned.
- 1 min: production-readiness gaps and next steps.

Capstone 2 — Intelligent IT Incident Triage & Resolution

IT triage agent + RAG + REST API/custom connector + agent flow + optional MCP/specialist hand-off

1. Business scenario

An IT service desk receives repetitive VPN, Wi-Fi, Microsoft 365, device, and access incidents. Build an agent that collects relevant details, retrieves approved troubleshooting guidance, suggests priority, and creates a ticket through an authorized API or agent flow. It must handle uncertain answers, integration failures, security-sensitive requests, and escalation.

2. Business goals

- Reduce repetitive first-line troubleshooting effort.
- Collect consistent incident details before escalation.
- Use grounded retrieval rather than unsupported troubleshooting claims.
- Demonstrate a ticket API/flow with validation and error handling.
- Measure routing, resolution guidance, safety, and operational readiness.

3. Personas

- Employee — reports the issue and answers diagnostic questions.
- IT Triage Agent — intake, retrieval, classification, actions, escalation.
- Knowledge/RAG source — runbooks, known errors, IT FAQs.
- Ticket action — authorized connector/flow or mock API.
- Human technician — handles complex, sensitive, or unresolved cases.
- Platform owner — manages DLP, roles, connections, tests, deployment.

4. Proposed architecture

- Employee in Teams/web → IT Triage Agent in Copilot Studio.
- Use generative orchestration or explicit topics for intake, slot filling, conditions, severity, fallback, and escalation.
- Approved runbooks and FAQs as knowledge; optionally Azure AI Search RAG with citations.
- Custom connector/REST API or agent flow creates a ticket; mock endpoint is acceptable for training.
- Optional scoped read-only MCP tool and/or identity/Microsoft 365 specialist-agent hand-off.
- Dev/Test/Prod, environment variables, connection references, DLP, least privilege, managed-solution deployment.

5. Functional and technical requirements

Area	Requirements
Incident intake	Collect service, symptoms, start time, scope, business impact, error text, device/network context, and attempted fixes; do not ask again for information already supplied.
Triage	Suggest priority using a documented sample matrix; state that final severity may require human confirmation. Route security incidents through approved escalation.
Knowledge/RAG	Use approved runbooks and cite sources where supported. If no relevant source is found, say so and escalate rather than inventing a fix.
Integration	Validate required fields and call ticket flow/API; display ticket ID only after confirmed success. Handle timeout, 4xx/5xx, invalid input, and duplicate submission.
MCP/multi-agent extension	Optionally add one safe, scoped MCP tool or specialist-agent hand-off; document permissions and test failures.
Governance/operations	Apply DLP and role controls; never request passwords/MFA codes; test, review analytics, publish to a test channel, and document rollback.

6. Implementation plan

Phase	Activities
1 — Plan & configure	Define incident scope, categories, sample severity matrix, privacy boundaries, environment strategy, DLP, roles, and solution structure.
2 — Build conversation	Create intake/troubleshooting topics or instructions; use variables/slot filling, conditions, fallback, adaptive card if useful, and escalation.
3 — Knowledge/RAG	Add approved runbooks and FAQs; test relevant, irrelevant, conflicting, and missing-answer cases.
4 — Integrate & extend	Build ticket flow/API; optionally add MCP or specialist hand-off; validate inputs and handle failures.
5 — Test & manage	Test classification, slots, retrieval, severity, API errors, privacy, fallback, escalation, analytics, and deployment.
6 — Deploy/demo	Publish to Teams/web test channel; package solution; document variables, connection references, promotion, monitoring, and rollback.

7. Demo / test scenarios

Scenario	Expected behavior
Routine incident	"Wi-Fi connects but internal sites do not open; it started 20 minutes ago." Expected: gather missing context and retrieve the approved runbook.
Create ticket	"Create a ticket: VPN disconnects every five minutes; started this morning; affects only me." Show ticket ID only if the API/flow confirms success.
Security incident	"I opened a suspicious attachment and entered my password." Follow approved security escalation; do not request the password.
Missing/secret knowledge	Ask for an internal emergency recovery code. Do not disclose or invent secrets; route to authorized procedures.
API failure	Simulate timeout or HTTP 500. Explain ticket creation could not be confirmed and provide manual fallback without blindly duplicating the request.
Specialist hand-off	"I can sign in to Windows but Microsoft 365 keeps blocking me." Hand off to configured identity/M365 specialist or authorized support.

8. Deliverables

- Incident-triage blueprint and sample severity matrix.
- IT Triage Agent with intake, troubleshooting, fallback, and escalation.
- Knowledge collection and retrieval test results.
- Ticket-creation flow/API, preferably tested against a mock service if needed.
- Optional MCP tool and/or specialist-agent hand-off.
- DLP, security, ALM, monitoring, rollback notes.
- Test matrix covering success, invalid input, timeout, API error, privacy, escalation.
- 8–10 minute demo and operations handover.

9. Acceptance criteria

- Agent captures required incident fields without repeated questions.
- Grounded troubleshooting uses approved sources; citations appear where supported.
- Ticket ID appears only after confirmed tool success.
- Security-sensitive incidents follow defined escalation.
- At least 10/12 non-critical tests pass; all privacy/security tests pass.
- Deployment, environment configuration, monitoring, and rollback are documented.

10. AB-620 domain mapping

Exam domain	Project evidence
Plan & configure (30–35%)	Use case, conversational design, environments, DLP, roles, solution structure, ALM, connections.
Integrate & extend (40–45%)	REST/custom connector, agent flow, optional MCP, specialist hand-off, RAG, validation/error handling.
Test & manage (20–25%)	Test matrix, analytics, Teams/web deployment, managed solution/pipeline, monitoring/maintenance.

Suggested guided build: 3–5 hours, with optional extension time. This is a training estimate, not an exam requirement.

Suggested 10-minute presentation

- 1 min: problem, users, success measures.
- 2 min: architecture, agent roles, knowledge, and governance.
- 4 min: end-to-end demo including a failure/escalation.
- 2 min: tests, analytics, deployment, lessons learned.
- 1 min: production-readiness gaps and next steps.

Shared scoring rubric (100 points)

Criterion	Evidence	Points
Planning, architecture, governance	Use case, environments, DLP, security, ALM	20
Agent and conversation design	Instructions/topics, orchestration, variables, fallback	15
Knowledge and RAG	Sources, grounding, citations where supported	15
Integration and collaboration	API/connector/flow/MCP/A2A or connected agents; error handling	25
Testing and operations	Test coverage, analytics, channel, deployment, monitoring	20
Demo and documentation	Walkthrough, design decisions, limitations, handover	5

Suggested instructor benchmark: 75/100 with no critical privacy/security failure. This is not an official AB-620 passing standard.

Submission checklist

- Architecture diagram and design decisions.
- Agent instructions, tool descriptions, knowledge-source inventory.
- Environment, DLP, roles, connection references, and ALM notes.
- Test cases with expected/actual results and fixes.
- Successful demo and one safe failure path.
- Deployment, monitoring, maintenance, and rollback notes.